

Lab 10

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Roll no : 38

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Section : A

Task 1

```
l1@DESKTOP-KQE4P80 MINGW64 ~ (main)
winget install --id GitHub.cli
```

```
: First copy your one-time code: 0A50 C194
Press Enter to open https://github.com/login/device in your browser
✓ Authentication complete.
- gh config set -h github.com git_protocol https
✓ Configured git protocol
✓ Logged in as 23-22412-038-sys
! You were already logged in to this account
```

```
$ gh codespace list
```

NAME	DISPLAY NAME
sturdy-parake...	sturdy par...
obscure-zebra...	obscure zebra
reimagined-ba...	reimagined...
ominous-waffl...	ominous wa...

```
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-1030-azure x86_64)

* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:       https://ubuntu.com/pro

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.
```

Task 2

```

inflating: aws/dist/prompts/LOCAL-3.0.51.dist-info/licenses/LICENSE
inflating: aws/dist/wheel-0.45.1.dist-info/direct_url.json
inflating: aws/dist/wheel-0.45.1.dist-info/METADATA
inflating: aws/dist/wheel-0.45.1.dist-info/INSTALLER
inflating: aws/dist/wheel-0.45.1.dist-info/entry_points.txt
inflating: aws/dist/wheel-0.45.1.dist-info/WHEEL
inflating: aws/dist/wheel-0.45.1.dist-info/LICENSE.txt
inflating: aws/dist/wheel-0.45.1.dist-info/RECORD
inflating: aws/dist/wheel-0.45.1.dist-info/REQUESTED
You can now run: /usr/local/bin/aws --version
aws-cli/2.32.24 Python/3.13.11 Linux/6.8.0-1030-azure exe/x86_64.ubuntu.24

```

```

cat ~/.aws/config
[default]
aws_access_key_id = AKIAZCN5ZS5DATBPGP50
aws_secret_access_key = N5eC7zbrdor1vNf3KxUHFMc150p/d7VP7Q9s3Vgy
[default]
region = me-central-1
output = json

```

```

{
  "UserId": "623705167686",
  "Account": "623705167686",
  "Arn": "arn:aws:iam::623705167686:root"
}

```

```

yrling.gpg
echo "deb [arch=$(dpkg --print-architecture) signed-by=/usr/share/keyrings/hashicorp-archive-keyring.gpg] \
https://apt.releases.hashicorp.com $(lsb_release -cs) main" | \
sudo tee /etc/apt/sources.list.d/hashicorp.list

sudo apt update
sudo apt install terraform
which terraform
terraform --version
--2025-12-28 13:00:41-- https://apt.releases.hashicorp.com/gpg
Resolving apt.releases.hashicorp.com [apt.releases.hashicorp.com]... 18.172.78.50, 18.172.78.65, 18.172.78.12, ...
Connecting to apt.releases.hashicorp.com [apt.releases.hashicorp.com] (18.172.78.50):443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 3980 (3.9K) [binary/octet-stream]
Saving to: 'STDOUT'

- 100%[=====] 3.99K --.-KB/s in 0s

2025-12-28 13:00:41 (958 MB/s) = written to stdout [3980/3980]

deb [arch=amd64 signed-by=/usr/share/keyrings/hashicorp-archive-keyring.gpg] https://apt.releases.hashicorp.com noble main
Get:1 https://apt.releases.hashicorp.com noble InRelease [12.9 kB]
Get:2 https://apt.releases.hashicorp.com noble/main amd64 Packages [266 kB]
Get:3 https://dl.yarnpkg.com/debian stable InRelease
Get:4 https://repo.anaconda.com/pkgs/misc/debrapo/conda stable InRelease [3961 B]
Get:5 https://packages.microsoft.com/repos/microsoft-ubuntu-noble-prod noble InRelease [3600 B]
Get:6 https://repo.anaconda.com/pkgs/misc/debrapo/conda stable/main amd64 Packages [4557 B]
Get:7 https://packages.microsoft.com/repos/microsoft-ubuntu-noble-prod noble/main all Packages [848 B]
Get:8 https://packages.microsoft.com/repos/microsoft-ubuntu-noble-prod noble/main amd64 Packages [77.3 kB]
Get:9 https://dl.yarnpkg.com/debian stable/main amd64 Packages [31.6 kB]
Get:10 https://dl.yarnpkg.com/debian stable/main all Packages [11.8 kB]
Get:11 https://archive.ubuntu.com/ubuntu noble InRelease [256 kB]
Get:12 https://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:13 https://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:14 https://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1732 kB]
Get:15 https://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:16 https://archive.ubuntu.com/ubuntu noble/multiverse amd64 Packages [331 kB]
Get:17 https://archive.ubuntu.com/ubuntu noble/universe amd64 Packages [39.3 MB]
Get:18 https://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Packages [33.1 kB]
Get:19 https://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1183 kB]

```


Initializing the backend...

Initializing provider plugins...

- Finding latest version of hashicorp/aws...

- Installing hashicorp/aws v6.27.0...

- Installed hashicorp/aws v6.27.0 (signed by HashiCorp)

Terraform has created a lock file **.terraform.lock.hcl** to record the provider selections it made above. Include this file in your version control repository so that Terraform can guarantee to make the same selections by default when you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see any changes that are required for your infrastructure. All Terraform commands should now work.

If you ever set or change modules or backend configuration for Terraform, rerun this command to reinitialize your working directory. If you forget, other commands will detect it and remind you to do so if necessary.

```
ls .terraform/
# This file is maintained automatically by "terraform init".
# Manual edits may be lost in future updates.

provider "registry.terraform.io/hashicorp/aws" {
  version = "6.27.0"
  hashes = [
    "h1:bixp2PSsP5ZG8czGCxcbsDn6lF5QF1UX1Nroq9cdab4=",
    "zh:177a24b806c72e8484b5cab93b2b38e3d770ae6f745a998b54d6619fd0e8129",
    "zh:4ac4a85c14fb868a3306b542e6a56c10bd6c6d5a67bc0c9b8f6a9060cf5f3be7",
    "zh:552652185bc85c8ba1da1d65dea47c454728a5c6839c458b6dcd3ce71c19ccfc",
    "zh:60284b8172d09aee91eae0856f09855eaf040ce3a58d6933602ae17c53f8ed04",
    "zh:6be38d156756ca61fb8e7c752cc5d769cd709686700ac4b230f40a6e95b5dbc9",
    "zh:7a409138fae4ef42e3a637e37cb9efedf96459e28a3c764fc4e855e8db9a7485",
    "zh:8070cf5224ed1ed3a3e9a59f7c30ff88bf071c7567165275d477c1738a56c064",
    "zh:894439ef340a9a79f69cd759e27ad11c7826adeca27be1b1ca82b3c9702fa300",
    "zh:89d035eebf08a97c89374ff06040955ddc09f275ecca609d0c9d58d149bef5cf",
    "zh:985b1145d724fc1f38369099e4a5087141885740fd6c0b1dbc492171e73c2e49",
    "zh:9b12af85486a96aedd8d7984b0ff811a4b42e3d88dad1a3fb4c0b580d04fa425",
    "zh:a80b47ae8d1475201c86bd94a5dcb9dd4da5e8b73102a90820b68b66b76d50fd",
    "zh:d3395be1556210f82199b9166a6b2e677cee9c4b67e96e63f6c3a98325ad7ab0",
    "zh:db0b869d09657f6f1e4110b56093c5fcd9dbdd97c020db1e577b239c0adcbce",
    "zh:ffc72e680370ae7c21f9bd3082c6317730df805c6797427839a6b6b7e9a26a01",
  ]
}
```

Task 3

```

provider "aws" {
  shared_config_files    = ["~/.aws/config"]
  shared_credentials_files = ["~/.aws/credentials"]
}

resource "aws_vpc" "development_vpc" {
  cidr_block = "10.0.0.0/16"
}

resource "aws_subnet" "dev_subnet_1" {
  vpc_id            = aws_vpc.development_vpc.id
  cidr_block        = "10.0.10.0/24"
  availability_zone = "me-central-1a"
}

```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:

- + create

Terraform will perform the following actions:

```

# aws_subnet.dev_subnet_1 will be created
+ resource "aws_subnet" "dev_subnet_1" {
  + arn                                = (known after apply)
  + assign_ipv6_address_on_creation    = false
  + availability_zone                  = "me-central-1a"
  + availability_zone_id               = (known after apply)
  + cidr_block                        = "10.0.10.0/24"
  + enable_dns64                      = false
  + enable_resource_name_dns_a_record_on_launch = false
  + enable_resource_name_dns_aaaa_record_on_launch = false
  + id                                = (known after apply)
  + ipv6_cidr_block_association_id     = (known after apply)
  + ipv6_native                        = false
  + map_public_ip_on_launch           = false
  + owner_id                          = (known after apply)
  + private_dns_hostname_type_on_launch = (known after apply)
  + region                            = "me-central-1"
  + tags_all                          = (known after apply)
  + vpc_id                            = (known after apply)
}

# aws_vpc.development_vpc will be created
+ resource "aws_vpc" "development_vpc" {
  + arn                                = (known after apply)
  + cidr_block                        = "10.0.0.0/16"
  + default_network_acl_id            = (known after apply)
  + default_route_table_id           = (known after apply)
  + default_security_group_id         = (known after apply)
}

```

Enter a value: yes

```

aws_vpc.development_vpc: Creating...
aws_vpc.development_vpc: Creation complete after 2s [id=vpc-00e832dc4b5877ee2]
aws_subnet.dev_subnet_1: Creating...
aws_subnet.dev_subnet_1: Creation complete after 0s [id=subnet-0132ca84323b35596]

Apply complete! Resources: 2 added, 0 changed, 0 destroyed.

```

```
{
  "Subnets": [
    {
      "AvailabilityZoneId": "me1-az1",
      "MapCustomerOwnedIpOnLaunch": false,
      "OwnerId": "623705167686",
      "AssignIpv6AddressOnCreation": false,
      "Ipv6CidrBlockAssociationSet": [],
      "SubnetArn": "arn:aws:ec2:me-central-1:623705167686:subnet/subnet-0132ca84323b35596",
      "EnableDns64": false,
      "Ipv6Native": false,
      "PrivateDnsNameOptionsOnLaunch": {
        "HostnameType": "ip-name",
        "EnableResourceNameDnsARecord": false,
        "EnableResourceNameDnsAAAARecord": false
      },
      "BlockPublicAccessStates": {
        "InternetGatewayBlockMode": "off"
      },
      "SubnetId": "subnet-0132ca84323b35596",
      "State": "available",
      "VpcId": "vpc-00e832dc4b5877ee2",
      "CidrBlock": "10.0.10.0/24",
      "AvailableIpAddressCount": 251,
      "AvailabilityZone": "me-central-1a",
      "DefaultForAz": false,
      "MapPublicIpOnLaunch": false
    }
  ]
}
```

```
{
  "Vpcs": [
    {
      "OwnerId": "623705167686",
      "InstanceTenancy": "default",
      "CidrBlockAssociationSet": [
        {
          "AssociationId": "vpc-cidr-assoc-0c45e37cf75768225",
          "CidrBlock": "10.0.0.0/16",
          "CidrBlockState": {
            "State": "associated"
          }
        }
      ],
      "IsDefault": false,
      "BlockPublicAccessStates": {
        "InternetGatewayBlockMode": "off"
      },
      "VpcId": "vpc-00e832dc4b5877ee2",
      "State": "available",
      "CidrBlock": "10.0.0.0/16",
      "DhcpOptionsId": "dopt-02dd4ed017908b331"
    }
  ]
}
```

Task 4

```

provider "aws" {
  shared_config_files = ["~/.aws/config"]
  shared_credentials_files = ["~/.aws/credentials"]
}

resource "aws_vpc" "development_vpc" {
  cidr_block = "10.0.0.0/16"
}

resource "aws_subnet" "dev_subnet_1" {
  vpc_id      = aws_vpc.development_vpc.id
  cidr_block  = "10.0.10.0/24"
  availability_zone = "me-central-1a"
}

data "aws_vpc" "existing_vpc" {
  default = true
}

resource "aws_subnet" "dev_subnet_1_existing" {
  vpc_id      = data.aws_vpc.existing_vpc.id
  cidr_block  = "172.31.48.0/24"
  availability_zone = "me-central-1a"
}

```

```

data.aws_vpc.existing_vpc: Reading...
aws_vpc.development_vpc: Refreshing state... [id=vpc-00e832dc4b5877ee2]
data.aws_vpc.existing_vpc: Read complete after 0s [id=vpc-079a26ed9b9cb32f2]
aws_subnet.dev_subnet_1: Refreshing state... [id=subnet-0132ca8432b35596]

```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:

- + create

Terraform will perform the following actions:

```

# aws_subnet.dev_subnet_1_existing will be created
+ resource "aws_subnet" "dev_subnet_1_existing" {
  arn                               = (known after apply)
  assign_ipv6_address_on_creation   = false
  availability_zone                 = "me-central-1a"
  availability_zone_id              = (known after apply)
  cidr_block                       = "172.31.48.0/24"
  enable_dns64                     = false
  enable_resource_name_dns_a_record_on_launch = false
  enable_resource_name_dns_aaaa_record_on_launch = false
  id                               = (known after apply)
  ipv6_cidr_block_association_id    = (known after apply)
  ipv6_native                       = false
  map_public_ip_on_launch           = false
  owner_id                         = (known after apply)
  private_dns_hostname_type_on_launch = (known after apply)
  region                           = "me-central-1"
  tags_all                         = (known after apply)
  vpc_id                           = "vpc-079a26ed9b9cb32f2"
}

```

Plan: 1 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?

Terraform will perform the actions described above.
only 'yes' will be accepted to approve.

Enter a value: yes

aws_subnet.dev_subnet_1_existing: Creating...

aws_subnet.dev_subnet_1_existing: Creation complete after 1s [id=subnet-07f2eeb9f45745f76]

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.


```

terraform refresh
terraform apply
terraform destroy
terraform plan
terraform apply
data.aws_vpc.existing_vpc: Reading...
data.aws_vpc.existing_vpc: Read complete after 0s [id=vpc-079a26ed9b9cb32f2]
aws_subnet.dev_subnet_1_existing: Refreshing state... [id=subnet-07f2eeb9f45745f76]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
- destroy

Terraform will perform the following actions:

# aws_subnet.dev_subnet_1_existing will be destroyed
- resource "aws_subnet" "dev_subnet_1_existing" {
  - arn                                     = "arn:aws:ec2:me-central-1:623705167686:subnet/subnet-07f2eeb9f45745f76" -> null
  - assign_ipv6_address_on_creation        = false -> null
  - availability_zone                     = "me-central-1a" -> null
  - availability_zone_id                  = "mec1-az1" -> null
  - cidr_block                            = "172.31.48.0/24" -> null
  - enable_dns64                          = false -> null
  - enable_inet_at_device_index           = 0 -> null
  - enable_resource_name_dns_a_record_on_launch = false -> null
  - enable_resource_name_dns_aaaa_record_on_launch = false -> null
  - id                                    = "subnet-07f2eeb9f45745f76" -> null
  - ipv6_native                           = false -> null
  - map_customer_owned_ip_on_launch       = false -> null
  - map_public_ip_on_launch               = false -> null
  - owner_id                              = "623705167686" -> null
  - private_dns_hostname_type_on_launch   = "ip-name" -> null
  - region                                = "me-central-1" -> null
  - tags                                  = {} -> null
  - tags_all                              = {} -> null
}

```

```

Destroy complete! Resources: 1 destroyed.
data.aws_vpc.existing_vpc: Reading...
aws_vpc.development_vpc: Refreshing state... [id=vpc-00e832dc4b5877ee2]
data.aws_vpc.existing_vpc: Read complete after 1s [id=vpc-079a26ed9b9cb32f2]
aws_subnet.dev_subnet_1: Refreshing state... [id=subnet-0132ca84323b35596]
data.aws_vpc.existing_vpc: Reading...
aws_vpc.development_vpc: Refreshing state... [id=vpc-00e832dc4b5877ee2]
data.aws_vpc.existing_vpc: Read complete after 1s [id=vpc-079a26ed9b9cb32f2]
aws_subnet.dev_subnet_1: Refreshing state... [id=subnet-0132ca84323b35596]

```

```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create

Terraform will perform the following actions:

# aws_subnet.dev_subnet_1_existing will be created
+ resource "aws_subnet" "dev_subnet_1_existing" {
  + arn                                     = (known after apply)
  + assign_ipv6_address_on_creation        = false
  + availability_zone                     = "me-central-1a"
  + availability_zone_id                  = (known after apply)
  + cidr_block                            = "172.31.48.0/24"
  + enable_dns64                          = false
  + enable_resource_name_dns_a_record_on_launch = false
  + enable_resource_name_dns_aaaa_record_on_launch = false
  + id                                    = (known after apply)
  + ipv6_cidr_block_association_id         = (known after apply)
  + ipv6_native                           = false
  + map_customer_owned_ip_on_launch       = false
  + map_public_ip_on_launch               = (known after apply)
  + owner_id                              = (known after apply)
  + private_dns_hostname_type_on_launch   = (known after apply)
  + region                                = "me-central-1"
  + tags                                  = (known after apply)
  + vpc_id                                = "vpc-079a26ed9b9cb32f2"
}

Plan: 1 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

  Enter a value: yes

aws_subnet.dev_subnet_1_existing: Creating...
aws_subnet.dev_subnet_1_existing: Creation complete after 0s [id=subnet-03a88faa503e156a]

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.
data.aws_vpc.existing_vpc: Reading...

```


Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
- destroy

Terraform will perform the following actions:

```
# aws_subnet.dev_subnet_1 will be destroyed
- resource "aws_subnet" "dev_subnet_1" {
  - arn                                = "arn:aws:ec2:me-central-1:623705167686:subnet/subnet-0132ca84323b35596" -> null
  - assign_ipv6_address_on_creation    = false -> null
  - availability_zone                  = "me-central-1a" -> null
  - availability_zone_id               = "mecl-az1" -> null
  - cidr_block                         = "10.0.10.0/24" -> null
  - enable_dns64                       = false -> null
  - enable_ini_at_device_index         = 0 -> null
  - enable_resource_name_dns_a_record_on_launch = false -> null
  - enable_resource_name_dns_aaaa_record_on_launch = false -> null
  - id                                 = "subnet-0132ca84323b35596" -> null
  - ipv6_native                        = false -> null
  - map_customer_owned_ip_on_launch    = false -> null
  - map_public_ip_on_launch            = false -> null
  - owner_id                           = "623705167686" -> null
  - private_dns_hostname_type_on_launch = "ip-name" -> null
  - region                             = "me-central-1" -> null
  - tags                               = {} -> null
  - tags_all                           = {} -> null
  - vpc_id                             = "vpc-00e832dc4b5877ee2" -> null
  # (4 unchanged attributes hidden)
}

# aws_subnet.dev_subnet_1_existing will be destroyed
- resource "aws_subnet" "dev_subnet_1_existing" {
  - arn                                = "arn:aws:ec2:me-central-1:623705167686:subnet/subnet-03a88faea503e156a" -> null
  - assign_ipv6_address_on_creation    = false -> null
  - availability_zone                  = "me-central-1a" -> null
  - availability_zone_id               = "mecl-az1" -> null
  - cidr_block                         = "172.31.48.0/24" -> null
  - enable_dns64                       = false -> null
  - enable_ini_at_device_index         = 0 -> null
  - enable_resource_name_dns_a_record_on_launch = false -> null
  - enable_resource_name_dns_aaaa_record_on_launch = false -> null
  - id                                 = "subnet-03a88faea503e156a" -> null
}

# aws_vpc.development_vpc will be destroyed
- resource "aws_vpc" "development_vpc" {
  - arn                                = "arn:aws:ec2:me-central-1:623705167686:vpc/vpc-00e832dc4b5877ee2" -> null
  - assign_generated_ipv6_cidr_block    = false -> null
  - cidr_block                         = "10.0.0.0/16" -> null
  - default_network_acl_id             = "acl-00182c8af491379a9" -> null
  - default_route_table_id             = "rtb-0ffeca4094c2af403" -> null
  - default_security_group_id          = "sg-0a7f615fdebccac670" -> null
  - dhcp_options_id                    = "dopt-02dd4ed017908b331" -> null
  - enable_dns_hostnames               = false -> null
  - enable_dns_support                  = true -> null
  - enable_network_address_usage_metrics = false -> null
  - id                                 = "vpc-00e832dc4b5877ee2" -> null
  - instance_tenancy                   = "default" -> null
  - ipv6_netmask_length                 = 0 -> null
  - main_route_table_id                = "rtb-0ffeca4094c2af403" -> null
  - owner_id                           = "623705167686" -> null
  - region                             = "me-central-1" -> null
  - tags                               = {} -> null
  - tags_all                           = {} -> null
  # (4 unchanged attributes hidden)
}
```

Plan: 0 to add, 0 to change, 3 to destroy.

Do you really want to destroy all resources?

Terraform will destroy all your managed infrastructure, as shown above.
There is no undo. Only 'yes' will be accepted to confirm.

Enter a value: yes

```
aws_subnet.dev_subnet_1_existing: Destroying... [id=subnet-03a88faea503e156a]
aws_subnet.dev_subnet_1: Destroying... [id=subnet-0132ca84323b35596]
aws_subnet.dev_subnet_1_existing: Destruction complete after 1s
aws_subnet.dev_subnet_1: Destruction complete after 1s
aws_vpc.development_vpc: Destroying... [id=vpc-00e832dc4b5877ee2]
aws_vpc.development_vpc: Destruction complete after 1s
```

Destroy complete! Resources: 3 destroyed.

```
data.aws_vpc.existing_vpc: Reading...
data.aws_vpc.existing_vpc: Read complete after 0s [id=vpc-079a26ed9b9cb32f2]
```

```

Destroy complete! Resources: 3 destroyed.
data.aws_vpc.existing_vpc: Reading...
data.aws_vpc.existing_vpc: Read complete after 0s [id=vpc-079a26ed9b9cb32f2]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create

```

Terraform will perform the following actions:

```

# aws_subnet.dev_subnet_1 will be created
+ resource "aws_subnet" "dev_subnet_1" {
  + arn                                = (known after apply)
  + assign_ipv6_address_on_creation    = false
  + availability_zone                  = "me-central-1a"
  + availability_zone_id                = (known after apply)
  + cidr_block                         = "10.0.10.0/24"
  + enable_dns64                       = false
  + enable_resource_name_dns_a_record_on_launch = false
  + enable_resource_name_dns_aaaa_record_on_launch = false
  + id                                 = (known after apply)
  + ipv6_cidr_block_association_id     = (known after apply)
  + ipv6_native                        = false
  + map_public_ip_on_launch            = false
  + owner_id                           = (known after apply)
  + private_dns_hostname_type_on_launch = (known after apply)
  + region                             = "me-central-1"
  + tags_all                           = (known after apply)
  + vpc_id                             = (known after apply)
}

# aws_subnet.dev_subnet_1_existing will be created
+ resource "aws_subnet" "dev_subnet_1_existing" {
  + arn                                = (known after apply)
  + assign_ipv6_address_on_creation    = false
  + availability_zone                  = "me-central-1a"
  + availability_zone_id                = (known after apply)
  + cidr_block                         = "172.31.48.0/24"
  + enable_dns64                       = false
  + enable_resource_name_dns_a_record_on_launch = false
  + enable_resource_name_dns_aaaa_record_on_launch = false
  + id                                 = (known after apply)
}

```

```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create

```

Terraform will perform the following actions:

```

# aws_subnet.dev_subnet_1 will be created
+ resource "aws_subnet" "dev_subnet_1" {
  + arn                                = (known after apply)
  + assign_ipv6_address_on_creation    = false
  + availability_zone                  = "me-central-1a"
  + availability_zone_id                = (known after apply)
  + cidr_block                         = "10.0.10.0/24"
  + enable_dns64                       = false
  + enable_resource_name_dns_a_record_on_launch = false
  + enable_resource_name_dns_aaaa_record_on_launch = false
  + id                                 = (known after apply)
  + ipv6_cidr_block_association_id     = (known after apply)
  + ipv6_native                        = false
  + map_public_ip_on_launch            = false
  + owner_id                           = (known after apply)
  + private_dns_hostname_type_on_launch = (known after apply)
  + region                             = "me-central-1"
  + tags_all                           = (known after apply)
  + vpc_id                             = (known after apply)
}

# aws_subnet.dev_subnet_1_existing will be created
+ resource "aws_subnet" "dev_subnet_1_existing" {
  + arn                                = (known after apply)
  + assign_ipv6_address_on_creation    = false
  + availability_zone                  = "me-central-1a"
  + availability_zone_id                = (known after apply)
  + cidr_block                         = "172.31.48.0/24"
  + enable_dns64                       = false
  + enable_resource_name_dns_a_record_on_launch = false
  + enable_resource_name_dns_aaaa_record_on_launch = false
  + id                                 = (known after apply)
  + ipv6_cidr_block_association_id     = (known after apply)
  + ipv6_native                        = false
  + map_public_ip_on_launch            = false
  + owner_id                           = (known after apply)
  + private_dns_hostname_type_on_launch = (known after apply)
  + region                             = "me-central-1"
}

```

Plan: 3 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?

Terraform will perform the actions described above.
Only 'yes' will be accepted to approve.

Enter a value: yes

aws_vpc.development_vpc: Creating...

aws_subnet.dev_subnet_1_existing: Creating...

aws_subnet.dev_subnet_1_existing: Creation complete after 1s [id=subnet-056d280c40cb85779]

aws_vpc.development_vpc: Creation complete after 1s [id=vpc-08bae3f872bd21535]

aws_subnet.dev_subnet_1: Creating...

aws_subnet.dev_subnet_1: Creation complete after 1s [id=subnet-02a9b9c4a3211c350]

Apply complete! Resources: 3 added, 0 changed, 0 destroyed.

```
provider "aws" {
  shared_config_files    = ["~/.aws/config"]
  shared_credentials_files = ["~/.aws/credentials"]
}

resource "aws_vpc" "development_vpc" {
  cidr_block = "10.0.0.0/16"
  tags = {
    Name      = "development"
    vpc_env   = "dev"
  }
}
```

terraform apply -auto-approve

aws_vpc.development_vpc: Refreshing state... [id=vpc-08bae3f872bd21535]

aws_subnet.dev_subnet_1: Refreshing state... [id=subnet-02a9b9c4a3211c350]

aws_subnet.dev_subnet_1_existing: Refreshing state... [id=subnet-056d280c40cb85779]

aws_vpc.development_vpc: Refreshing state... [id=vpc-08bae3f872bd21535]

aws_subnet.dev_subnet_1_existing: Refreshing state... [id=subnet-056d280c40cb85779]

aws_subnet.dev_subnet_1: Refreshing state... [id=subnet-02a9b9c4a3211c350]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
- update in-place
- destroy

Terraform will perform the following actions:

```
# aws_subnet.dev_subnet_1 will be destroyed
# (because aws_subnet.dev_subnet_1 is not in configuration)
- resource "aws_subnet" "dev_subnet_1" {
  - arn                                = "arn:aws:ec2:me-central-1:623705167686:subnet/subnet-02a9b9c4a3211c350" -> null
  - assign_ipv6_address_on_creation    = false -> null
  - availability_zone                  = "me-central-1a" -> null
  - availability_zone_id               = "mec1-az1" -> null
  - cidr_block                         = "10.0.10.0/24" -> null
  - enable_dns64                      = false -> null
  - enable_lni_at_device_index        = 0 -> null
  - enable_resource_name_dns_a_record_on_launch = false -> null
  - enable_resource_name_dns_aaaa_record_on_launch = false -> null
  - id                                = "subnet-02a9b9c4a3211c350" -> null
  - ipv6_native                       = false -> null
  - map_customer_owned_ip_on_launch    = false -> null
  - map_public_ip_on_launch           = false -> null
  - owner_id                          = "623705167686" -> null
  - private_dns_hostname_type_on_launch = "ip-name" -> null
  - region                            = "me-central-1" -> null
  - tags                              = {} -> null
  - tags_all                          = {} -> null
  - vpc_id                            = "vpc-08bae3f872bd21535" -> null
  # (4 unchanged attributes hidden)
}

# aws_subnet.dev_subnet_1_existing will be destroyed
# (because aws_subnet.dev_subnet_1_existing is not in configuration)
```

```

# (because aws_subnet.dev_subnet_1_existing is not in configuration)
- resource "aws_subnet" "dev_subnet_1_existing" {
  - arn = "arn:aws:ec2:me-central-1:623705167686:subnet-056d280c40cb85779"
  - assign_ipv6_address_on_creation = false -> null
  - availability_zone = "me-central-1a" -> null
  - availability_zone_id = "mec1-az1" -> null
  - cidr_block = "172.31.48.0/24" -> null
  - enable_dns64 = false -> null
  - enable_lni_at_device_index = 0 -> null
  - enable_resource_name_dns_a_record_on_launch = false -> null
  - enable_resource_name_dns_aaaa_record_on_launch = false -> null
  - id = "subnet-056d280c40cb85779" -> null
  - ipv6_native = false -> null
  - map_customer_owned_ip_on_launch = false -> null
  - map_public_ip_on_launch = false -> null
  - owner_id = "623705167686" -> null
  - private_dns_hostname_type_on_launch = "ip-name" -> null
  - region = "me-central-1" -> null
  - tags = {} -> null
  - tags_all = {} -> null
  - vpc_id = "vpc-079a26ed9b9cb32f2" -> null
  # (4 unchanged attributes hidden)
}

```

```

# aws_vpc.development_vpc will be updated in-place
- resource "aws_vpc" "development_vpc" {
  id = "vpc-08bae3f872bd21535"
  - tags = {
    + "Name" = "development"
    + "vpc_env" = "dev"
  }
  - tags_all = {
    + "Name" = "development"
    + "vpc_env" = "dev"
  }
  # (19 unchanged attributes hidden)
}

```

Plan: 0 to add, 1 to change, 2 to destroy.

aws_subnet.dev_subnet_1_existing: Destroying... [id=subnet-056d280c40cb85779]

aws_subnet.dev_subnet_1: Destroying... [id=subnet-02a9b9c4a3211c350]

aws_subnet.dev_subnet_1_existing: Destruction complete after 1s

aws_subnet.dev_subnet_1: Destruction complete after 1s

aws_vpc.development_vpc: Modifying... [id=vpc-08bae3f872bd21535]

aws_vpc.development_vpc: Modifications complete after 1s [id=vpc-08bae3f872bd21535]

Apply complete! Resources: 0 added, 1 changed, 2 destroyed.


```

terraform apply
aws_vpc.development_vpc: Refreshing state... [id=vpc-08bae3f872bd21535]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
~ update in-place

Terraform will perform the following actions:

# aws_vpc.development_vpc will be updated in-place
~ resource "aws_vpc" "development_vpc" {
  id          = "vpc-08bae3f872bd21535"
  ~ tags      = {
    "Name"      = "development"
    "vpc_env"   = "dev" -> null
  }
  ~ tags_all  = {
    "vpc_env" = "dev" -> null
    # (2 unchanged elements hidden)
  }
  # (19 unchanged attributes hidden)
}

Plan: 0 to add, 1 to change, 0 to destroy.

Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to take exactly these actions if you run "terraform apply" now.
aws_vpc.development_vpc: Refreshing state... [id=vpc-08bae3f872bd21535]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
~ update in-place

Terraform will perform the following actions:

# aws_vpc.development_vpc will be updated in-place
~ resource "aws_vpc" "development_vpc" {
  id          = "vpc-08bae3f872bd21535"
  ~ tags      = {
    "Name"      = "development"
    "vpc_env"   = "dev" -> null
  }
  ~ tags_all  = {
    "vpc_env" = "dev" -> null
    # (1 unchanged element hidden)
  }
  # (19 unchanged attributes hidden)
}

```

```

Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to take
aws_vpc.development_vpc: Refreshing state... [id=vpc-08bae3f872bd21535]

Terraform used the selected providers to generate the following execution plan. Resource ac
~ update in-place

Terraform will perform the following actions:

# aws_vpc.development_vpc will be updated in-place
~ resource "aws_vpc" "development_vpc" {
  id          = "vpc-08bae3f872bd21535"
  ~ tags      = {
    "Name"      = "development"
    "vpc_env"   = "dev" -> null
  }
  ~ tags_all  = {
    "vpc_env" = "dev" -> null
    # (1 unchanged element hidden)
  }
  # (19 unchanged attributes hidden)
}

Plan: 0 to add, 1 to change, 0 to destroy.

Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

  Enter a value: yes

aws_vpc.development_vpc: Modifying... [id=vpc-08bae3f872bd21535]
aws_vpc.development_vpc: Modifications complete after 0s [id=vpc-08bae3f872bd21535]

Apply complete! Resources: 0 added, 1 changed, 0 destroyed

```

Task 5

```

cat terraform.tfstate
cat terraform.tfstate.backup
terraform apply
cat terraform.tfstate
cat terraform.tfstate.backup
terraform state list
terraform state show aws_vpc.development_vpc
aws_vpc.development_vpc: Refreshing state... [id=vpc-08bae3f872bd21535]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
- destroy

Terraform will perform the following actions:

# aws_vpc.development_vpc will be destroyed
- resource "aws_vpc" "development_vpc" {
  - arn                                = "arn:aws:ec2:me-central-1:623705167686:vpc/vpc-08bae3f872bd21535" -> null
  - assign_generated_ipv6_cidr_block  = false -> null
  - cidr_block                        = "10.0.0.0/16" -> null
  - default_network_acl_id           = "acl-03c0a6ba5c9b37a31" -> null
  - default_route_table_id           = "rtb-01326304ae881fa22" -> null
  - default_security_group_id        = "sg-03094d41a61ebc8ef" -> null
  - dhcp_options_id                  = "dopt-02dd4ed017908b331" -> null
  - enable_dns_hostnames              = false -> null
  - enable_dns_support                = true -> null
  - enable_network_address_usage_metrics = false -> null
  - id                               = "vpc-08bae3f872bd21535" -> null
  - instance_tenancy                  = "default" -> null
  - ipv6_address_mask_length          = 0 -> null
  - main_route_table_id               = "rtb-01326304ae881fa22" -> null
  - owner_id                          = "623705167686" -> null
  - region                            = "me-central-1" -> null
  - tags                             = {
    - "Name" = "development"
  } -> null
  - tags_all                           = {
    - "Name" = "development"
  } -> null
  # (4 unchanged attributes hidden)
}

Plan: 0 to add, 0 to change, 1 to destroy.

Do you really want to destroy all resources?
Terraform will destroy all your managed infrastructure, as shown above.
There is no undo. Only 'yes' will be accepted to confirm.

```

```

Do you really want to destroy all resources?
Terraform will destroy all your managed infrastructure, as shown above.
There is no undo. Only 'yes' will be accepted to confirm.

```

Enter a value: yes

```

aws_vpc.development_vpc: Destroying... [id=vpc-08bae3f872bd21535]
aws_vpc.development_vpc: Destruction complete after 0s

```

Destroy complete! Resources: 1 destroyed.

```

{
  "version": 4,
  "terraform_version": "1.14.3",
  "serial": 27,
  "lineage": "3b3c750a-2506-d918-7465-edae5f5918f3",
  "outputs": {},
  "resources": [],
  "check_results": null
}
{
  "version": 4,
  "terraform_version": "1.14.3",
  "serial": 25,
  "lineage": "3b3c750a-2506-d918-7465-edae5f5918f3",
  "outputs": {},
  "resources": [
    {
      "mode": "managed",
      "type": "aws_vpc",
      "name": "development_vpc",
      "provider": "provider[\"registry.terraform.io/hashicorp/aws\"]",
      "instances": [
        {
          "schema_version": 1,
          "attributes": {
            "arn": "arn:aws:ec2:me-central-1:623705167686:vpc/vpc-08bae3f872bd21535",
            "assign_generated_ipv6_cidr_block": false,
            "cidr_block": "10.0.0.0/16",
            "default_network_acl_id": "acl-03c0a6ba5c9b37a31",
            "default_route_table_id": "rtb-01326304ae881fa22",
            "default_security_group_id": "sg-03094d41a61ebc8ef",
            "dhcp_options_id": "dopt-02dd4ed017908b331",
            "enable_dns_hostnames": false,
            "enable_dns_support": true,
            "enable_network_address_usage_metrics": false,
            "id": "vpc-08bae3f872bd21535",
            "instance_tenancy": "default",

```



```

+ ipv6_cidr_block_network_border_group = (known after apply)
+ main_route_table_id                  = (known after apply)
+ owner_id                             = (known after apply)
+ region                               = "me-central-1"
+ tags                                  = {
+   + "Name" = "development"
+ }
+ tags_all                              = {
+   + "Name" = "development"
+ }
}

```

Plan: 1 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?

Terraform will perform the actions described above.
Only 'yes' will be accepted to approve.

Enter a value: yes

aws_vpc.development_vpc: Creating...

aws_vpc.development_vpc: Creation complete after 2s [id=vpc-064ef4080514a164e]

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.

```

{
  "version": 4,
  "terraform_version": "1.14.3",
  "serial": 29,
  "lineage": "3b3c750a-2506-d918-7465-edae5f5918f3",
  "outputs": {},
  "resources": [
    {
      "mode": "managed",
      "type": "aws_vpc",
      "name": "development_vpc",
      "provider": "provider[\"registry.terraform.io/hashicorp/aws\"]",
      "instances": [
        {
          "schema_version": 1,
          "attributes": {
            "arn": "arn:aws:ec2:me-central-1:623705167686:vpc/vpc-064ef4080514a164e",
            "assign_generated_ipv6_cidr_block": false,
            "cidr_block": "10.0.0.0/16",
            "default_network_acl_id": "acl-0d19d823cb0f44c17",
            "default_route_table_id": "rtb-04b6a67794e4345da",
            "default_security_group_id": "sg-0a22cbdcaadb4ba55",
            "dhcp_options_id": "dopt-02dd4ed017908b331",
            "enable_dns_hostnames": false,

```


Task 6

Outputs :

```
dev-subnet-cidr_block = "10.0.10.0/24"
dev-subnet-region = "me-central-1a"
dev-subnet-tags_all = tomap({
  "Name" = "subnet-1-dev"
})
dev-subnet-tags_name = "subnet-1-dev"
dev-vpc-cidr_block = "10.0.0.0/16"
dev-vpc-region = "me-central-1"
dev-vpc-tags_all = tomap({
  "Name" = "development"
})
dev-vpc-tags_name = "development"
```

Clean up

```
cat terraform.tfstate
cat terraform.tfstate.backup
aws_vpc.development_vpc: Refreshing state... [id=vpc-064ef4080514a164e]
aws_subnet.dev_subnet_1: Refreshing state... [id=subnet-0af87dafd3b1dc4f4]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
- destroy

Terraform will perform the following actions:

# aws_subnet.dev_subnet_1 will be destroyed
- resource "aws_subnet" "dev_subnet_1" {
  - arn                                = "arn:aws:ec2:me-central-1:623705167686:subnet/subnet-0af87dafd3b1dc4f4" -> null
  - assign_ipv6_address_on_creation    = false -> null
  - availability_zone                  = "me-central-1a" -> null
  - availability_zone_id                = "mec1-az1" -> null
  - cidr_block                          = "10.0.10.0/24" -> null
  - enable_dns64                       = false -> null
  - enable_inflight_device_index       = 0 -> null
  - enable_resource_name_dns_a_record_on_launch = false -> null
  - enable_resource_name_dns_saaa_record_on_launch = false -> null
  - id                                 = "subnet-0af87dafd3b1dc4f4" -> null
  - ipv6_native                        = false -> null
  - map_customer_owned_ip_on_launch    = false -> null
  - map_public_ip_on_launch            = false -> null
  - owner_id                           = "623705167686" -> null
  - private_dns_hostname_type_on_launch = "ip-name" -> null
  - region                             = "me-central-1" -> null
  - tags                               = {
    - "Name" = "subnet-1-dev"
  } -> null
  - tags_all                           = {
    - "Name" = "subnet-1-dev"
  } -> null
  - vpc_id                             = "vpc-064ef4080514a164e" -> null
  # (4 unchanged attributes hidden)
}

# aws_vpc.development_vpc will be destroyed
```

```
# aws_vpc.development_vpc will be destroyed
- resource "aws_vpc" "development_vpc" {
  - arn = "arn:aws:ec2:me-central-1:623705167686:vpc/vpc-064ef4080514a164e" -> null
  - assign_generated_ipv6_cidr_block = false -> null
  - cidr_block = "10.0.0.0/16" -> null
  - default_network_acl_id = "acl-0d19d823cb0f44c17" -> null
  - default_route_table_id = "rtb-04b6a67794e4345da" -> null
  - default_security_group_id = "sg-0a22cbdcadb4ba55" -> null
  - dhcp_options_id = "dopt-02dd4ed017908b331" -> null
  - enable_dns_hostnames = false -> null
  - enable_dns_support = true -> null
  - enable_network_address_usage_metrics = false -> null
  - id = "vpc-064ef4080514a164e" -> null
  - instance_tenancy = "default" -> null
  - ipv6_netmask_length = 0 -> null
  - main_route_table_id = "rtb-04b6a67794e4345da" -> null
  - owner_id = "623705167686" -> null
  - region = "me-central-1" -> null
  - tags = {
    - "Name" = "development"
  } -> null
  - tags_all = {
    - "Name" = "development"
  } -> null
  # (4 unchanged attributes hidden)
}
```

Plan: 0 to add, 0 to change, 2 to destroy.

Changes to Outputs:

```
- dev-subnet-cidr_block = "10.0.10.0/24" -> null
- dev-subnet-region = "me-central-1a" -> null
- dev-subnet-tags_all = {
  - Name = "subnet-1-dev"
} -> null
- dev-subnet-tags_name = "subnet-1-dev" -> null
- dev-vpc-cidr_block = "10.0.0.0/16" -> null
- dev-vpc-region = "me-central-1" -> null
- dev-vpc-tags_all = {
  - Name = "development"
} -> null
- dev-vpc-tags_name = "development" -> null
```

Do you really want to destroy all resources?

Terraform will destroy all your managed infrastructure, as shown above.
There is no undo. Only 'yes' will be accepted to confirm.

Enter a value: yes

```
aws_subnet.dev_subnet_1: Destroying... [id=subnet-0af87dafd1b1dc4f4]
aws_subnet.dev_subnet_1: Destruction complete after 1s
aws_vpc.development_vpc: Destroying... [id=vpc-064ef4080514a164e]
aws_vpc.development_vpc: Destruction complete after 1s
```

Destroy complete! Resources: 2 destroyed.

```
{
  "version": 4,
  "terraform_version": "1.14.3",
  "serial": 34,
  "lineage": "3b3c750a-2506-d918-7465-edae5f5918f3",
  "outputs": {},
  "resources": [],
  "check_results": null
}
{
  "version": 4,
  "terraform_version": "1.14.3",
  "serial": 31,
  "lineage": "3b3c750a-2506-d918-7465-edae5f5918f3",
  "outputs": {
    "dev-subnet-cidr_block": {
      "value": "10.0.10.0/24",
      "type": "string"
    },
    "dev-subnet-region": {
      "value": "me-central-1a",
      "type": "string"
    },
    "dev-subnet-tags_all": {
      "value": {
        "Name": "subnet-1-dev"
      },
      "type": [
        "map",
        "string"
      ]
    },
    "dev-subnet-tags_name": {
      "value": "subnet-1-dev",
      "type": "string"
    },
    "dev-vpc-cidr_block": {
      "value": "10.0.0.0/16",
      "type": "string"
    },
    "dev-vpc-region": {
      "value": "me-central-1",

```



```

"type": "aws_vpc",
"name": "development_vpc",
"provider": "provider[\\registry.terraform.io/hashicorp/aws\\"]",
"instances": [
  {
    "schema_version": 1,
    "attributes": {
      "arn": "arn:aws:ec2:me-central-1:623705167686:vpc/vpc-064ef4080514a164e",
      "assign_generated_ipv6_cidr_block": false,
      "cidr_block": "10.0.0.0/16",
      "default_network_acl_id": "acl-0d19d823cb0f44c17",
      "default_route_table_id": "rtb-04b6a67794e4345da",
      "default_security_group_id": "sg-0a22cbdcaadb4ba55",
      "dhcp_options_id": "dopt-02dd4ed017908b331",
      "enable_dns_hostnames": false,
      "enable_dns_support": true,
      "enable_network_address_usage_metrics": false,
      "id": "vpc-064ef4080514a164e",
      "instance_tenancy": "default",
      "ipv4_ipam_pool_id": null,
      "ipv4_netmask_length": null,
      "ipv6_association_id": "",
      "ipv6_cidr_block": "",
      "ipv6_cidr_block_network_border_group": "",
      "ipv6_ipam_pool_id": "",
      "ipv6_netmask_length": 0,
      "main_route_table_id": "rtb-04b6a67794e4345da",
      "owner_id": "623705167686",
      "region": "me-central-1",
      "tags": {
        "Name": "development"
      },
      "tags_all": {
        "Name": "development"
      }
    },
    "sensitive_attributes": [],
    "identity_schema_version": 0,
    "identity": {
      "account_id": "623705167686",
      "id": "vpc-064ef4080514a164e",
      "region": "me-central-1"
    },
    "private": "eyJzY2h1bWFFdmVyc2lubiI6IjEifQ=="
  }
]
}

```

```

    },
    "sensitive_attributes": [],
    "identity_schema_version": 0,
    "identity": {
      "account_id": "623705167686",
      "id": "vpc-064ef4080514a164e",
      "region": "me-central-1"
    },
    "private": "eyJzY2h1bWFFdmVyc2lubiI6IjEifQ=="
  }
]
},
"check_results": null
}

```

